

IT TICKET ANALYSIS

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Problem Statement



Objective

Evaluate the IT support ticket system to measure agent performance, resolution speed, and employee satisfaction, and use these results to guide hiring, training, and exits to strengthen overall service quality.



What is IT Ticket Management

It is the end-to-end workflow for logging, classifying, prioritizing, assigning, tracking, and closing IT support requests, typically via help desk software, to deliver timely, consistent service and improve user experience.

Data Overview



- 97,498 tickets in 5 years.
- Number of Request Categories: 4
- Unique Categories: Login Access, System, Software, Hardware
- Number of Priorities: 4
- Unique Priorities: Unassigned, High, Mid, Low.
- Number of Agents: 50 agents
- Number of Severities: 5
- Unique Severities: Unclassified, Normal, Urgent, Major, Minor.

Data Cleaning and Preprocessing:

- Fixed spelling mistakes in Priority and Severity.
- Extracted the category from Priority and Severity.
- Added agent details with tickets
- Create a feature to group agents based on there Age.
- Extracted the age of each agent.
- Extracted domain name from agent email address.

Data Attributes

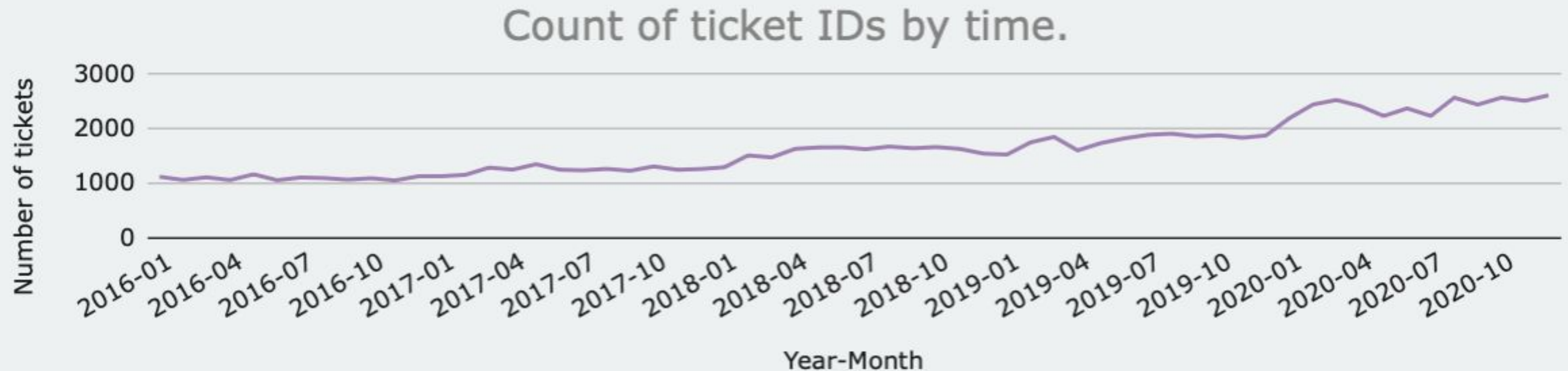
- **Tickets Sheet** : Contains information about IT support tickets.
- **ID Ticket**: Unique identifier for the ticket.
- **Fecha**: Date of the ticket.
- **Employee ID**: ID of the employee who raised the ticket.
- **Agent ID**: ID of the agent assigned to the ticket.
- **Request Category**: Category of the request (e.g., Login Access, System, Software).
- **Issue Type**: Type of issue (e.g., IT Error, IT Request).
- **Severity**: Severity of the issue.
- **Priority**: Priority level of the issue.
- **Resolution Time (Days)**: Time taken to resolve the ticket.
- **Satisfaction Rate**: Satisfaction rate provided by the employee (1 - 5 scale).
- **IT Agents**: Contains information about IT agents.
- **Agent ID**: Unique identifier for the agent.
- **Full Name**: Full name of the agent.
- **Email**: Email address of the agent.
- **Year of Birth**: Year the agent was born.
- **Month of Birth**: Month the agent was born.
- **Day of Birth**: Day the agent was born.

Data Analytical approach and tools



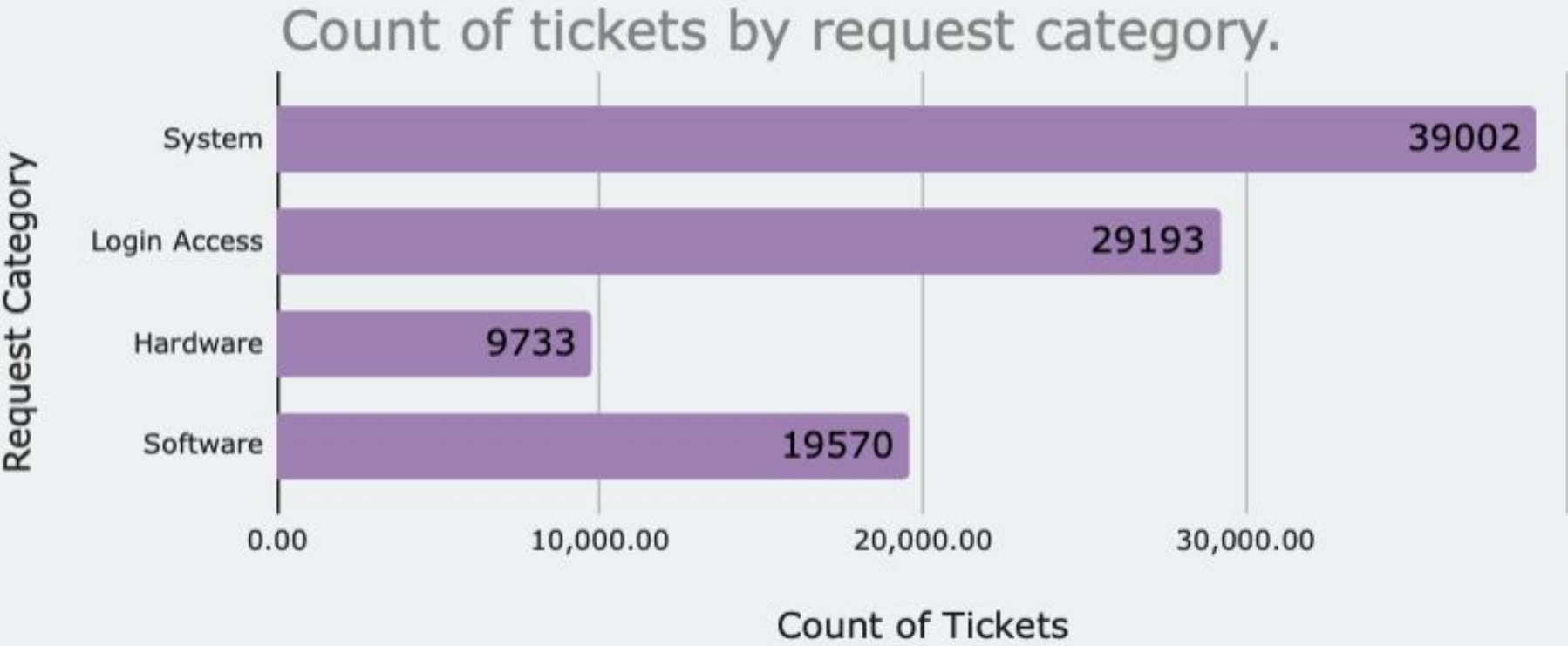
- Data Cleaning: Utilized functions **TRIM, LEN, FIND, RIGHT, ROUND**.
- Data Enrichment: Enhanced data set with additional variables using **VLOOKUP** to cross-reference external data source.
- Trend analysis: Conducted time series analysis using **PIVOT TABLE** and Charts to find trend in tickets and its behavior.
- Utilized **DATE, DATEDIF** to create feature columns in dataset.
- Visualization: Created dynamic charts and dashboard for data representation, enabling interactive data exploration.
- Filtering : Utilized **Pivot table** and **Slicers** feature to visualize and chronological filtering of data.
- Conducted **Cost Benefit Analysis** for investment decision making.
- Trend Analysis: Conducted **time series analysis** using line Chart

Count of ticked IDs by time



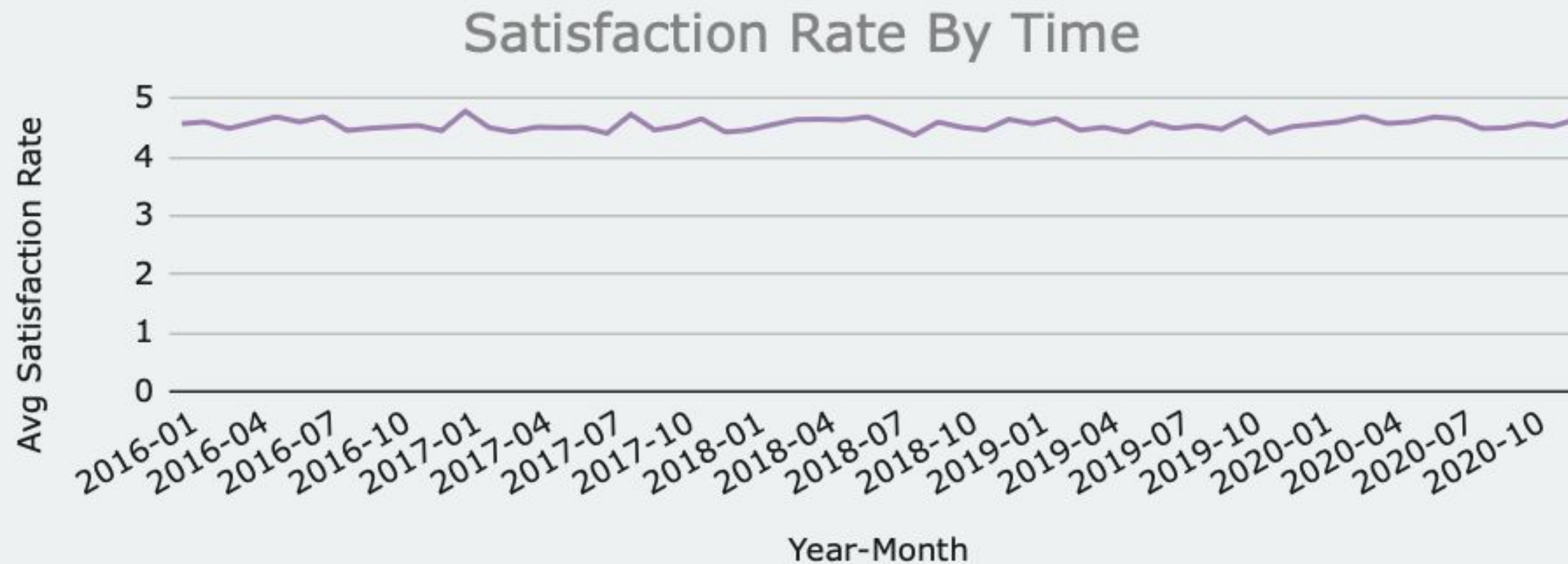
- Ticket volume has shown consistent and significant growth from 2016 to 2020, more than doubling. 2020 saw the most dramatic acceleration, with monthly tickets exceeding 2,000 and reaching 2,609 in December.
- Seasonal fluctuations exist with higher activity in Q1, mid-year, and year-end, but the overall trend is escalating demand.

Count of Tickets by Request Category



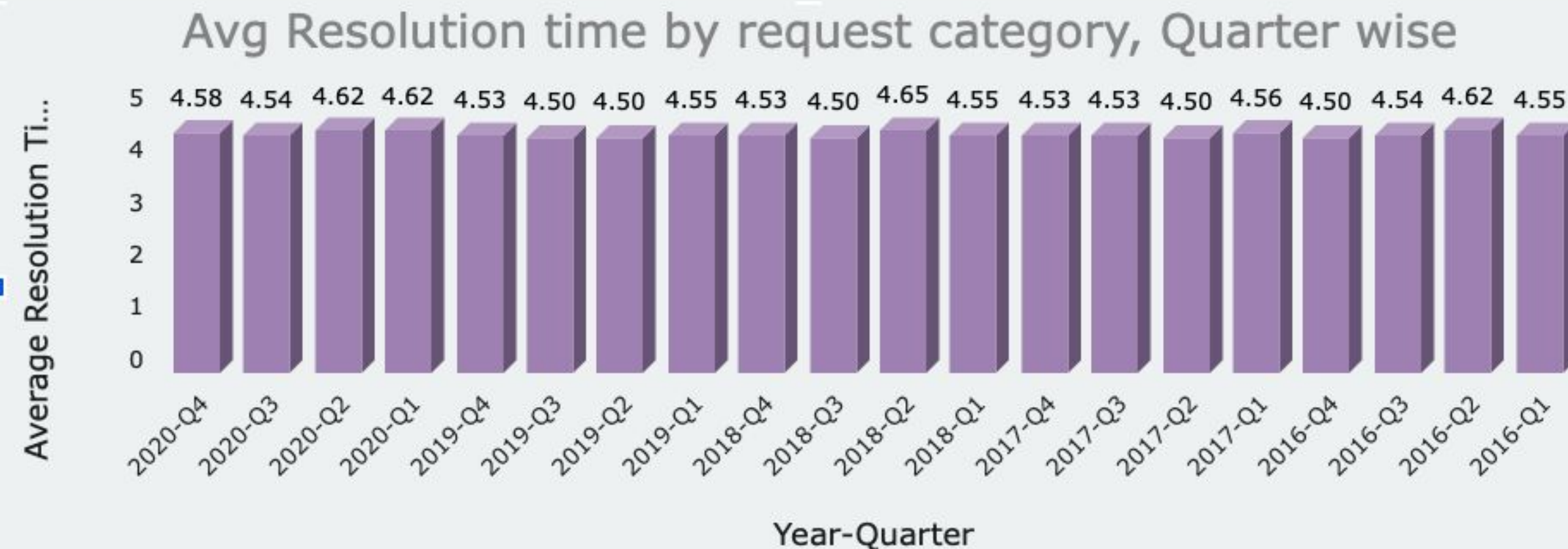
- System and Login Access categories account for the vast majority of IT requests.
- Hardware has the lowest ticket volume, but the longest resolution time.
- System issues account for 40% of tickets, Login Access 30%, Software 20%, and Hardware 10%

Satisfaction Rate by Time



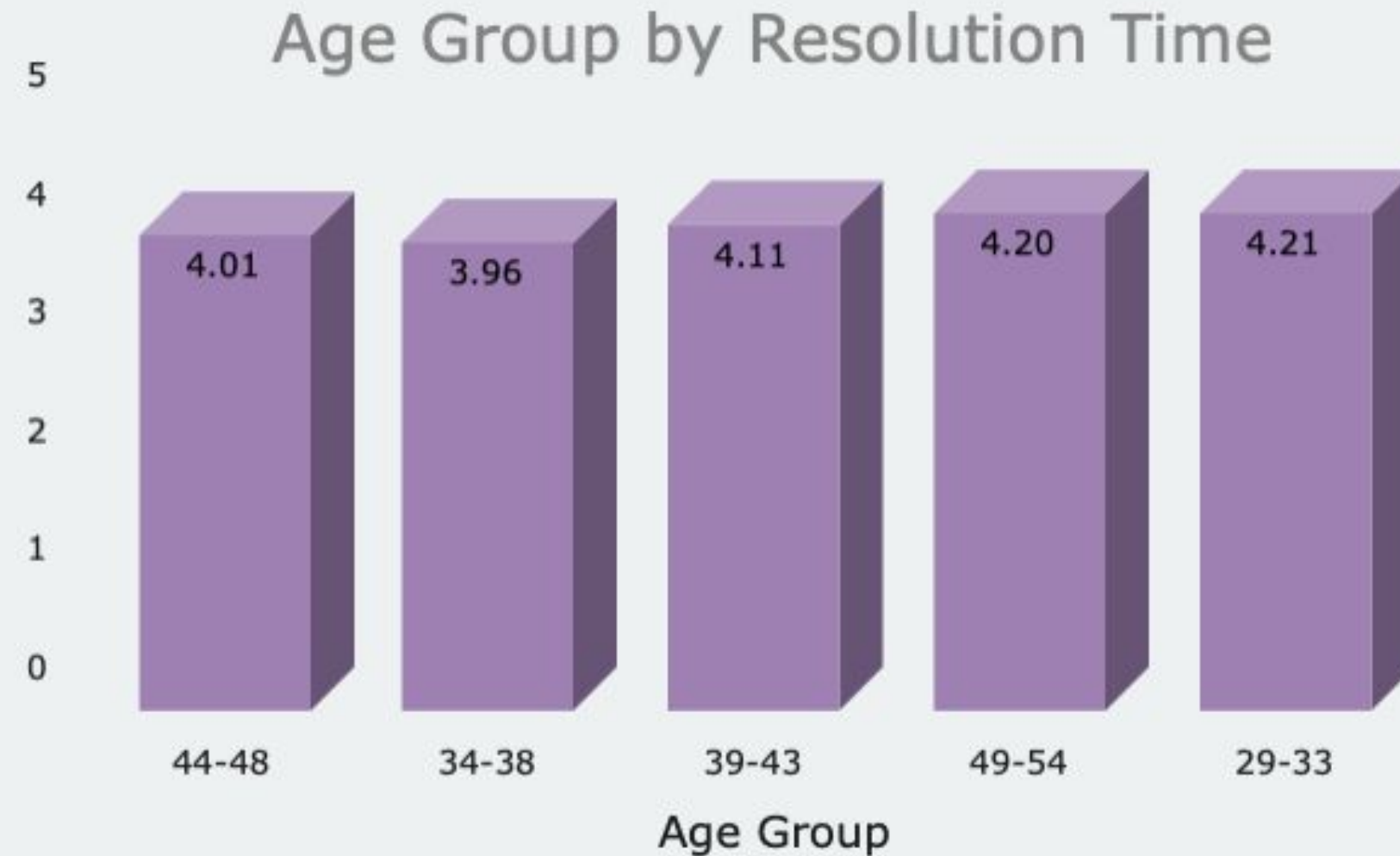
- Customer satisfaction remained consistently above 4.0, indicating the IT support team maintained a strong service experience despite fluctuations in ticket volume and resolution time.
- The stable trend suggests current service practices are effective, allowing the organization to focus future investments on improving efficiency rather than fixing customer experience.
- Since satisfaction has been maintained over multiple years, the organization has a strong foundation to introduce automation and process improvements with minimal risk to user experience.

Average resolution time by request category, quarter-wise



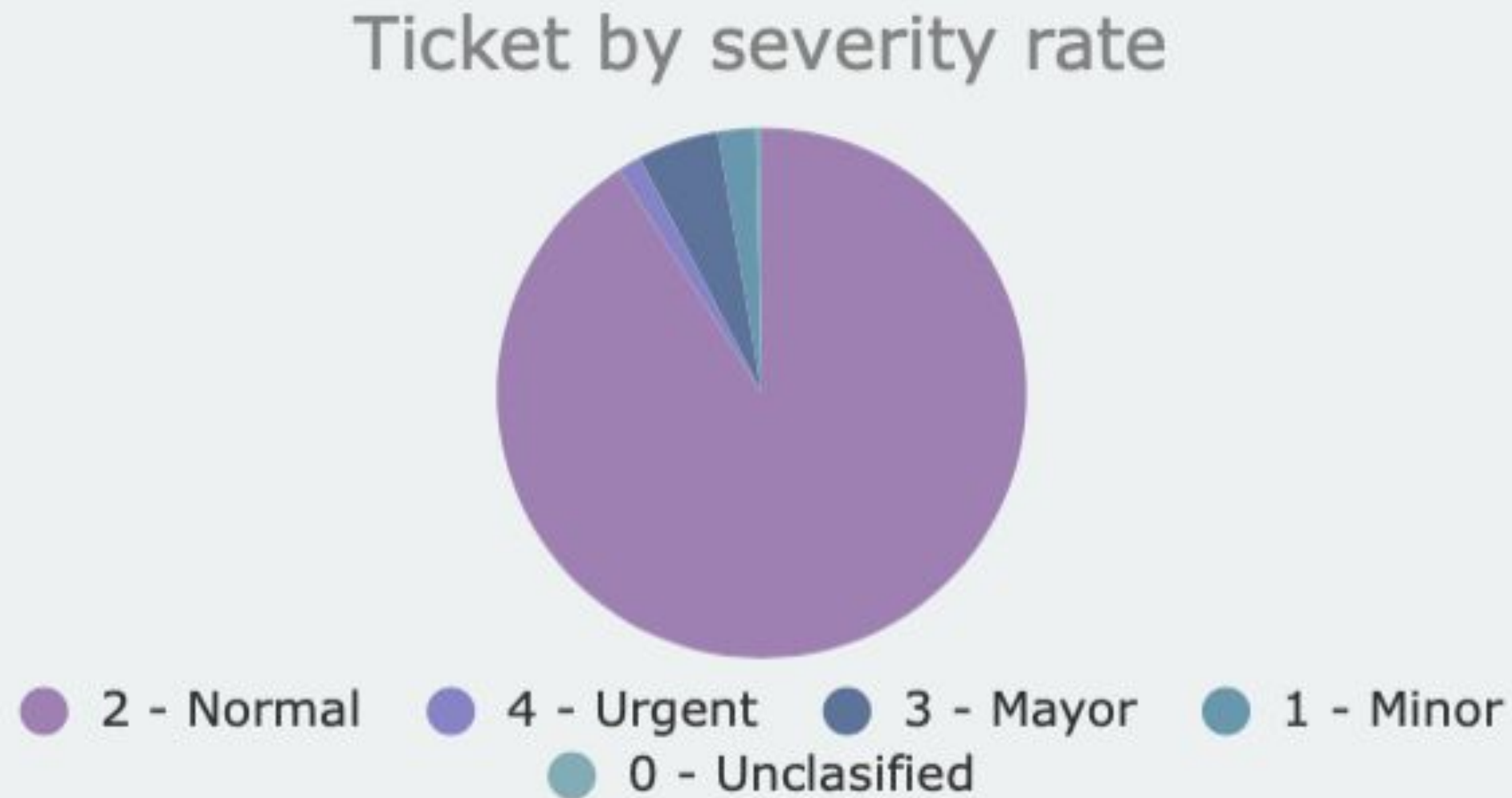
- Customer satisfaction is consistently high across all age groups, suggesting that service quality depends more on standardized processes than on employee age.
- Agents aged 49–54 achieved the highest satisfaction scores, indicating that experienced staff can play a valuable role in mentoring junior agents and handling complex tickets.
- Since performance differences are relatively small, the organization can maintain workforce diversity while focusing investments on training and knowledge sharing instead of age-specific hiring.

Age Group by Satisfaction rate



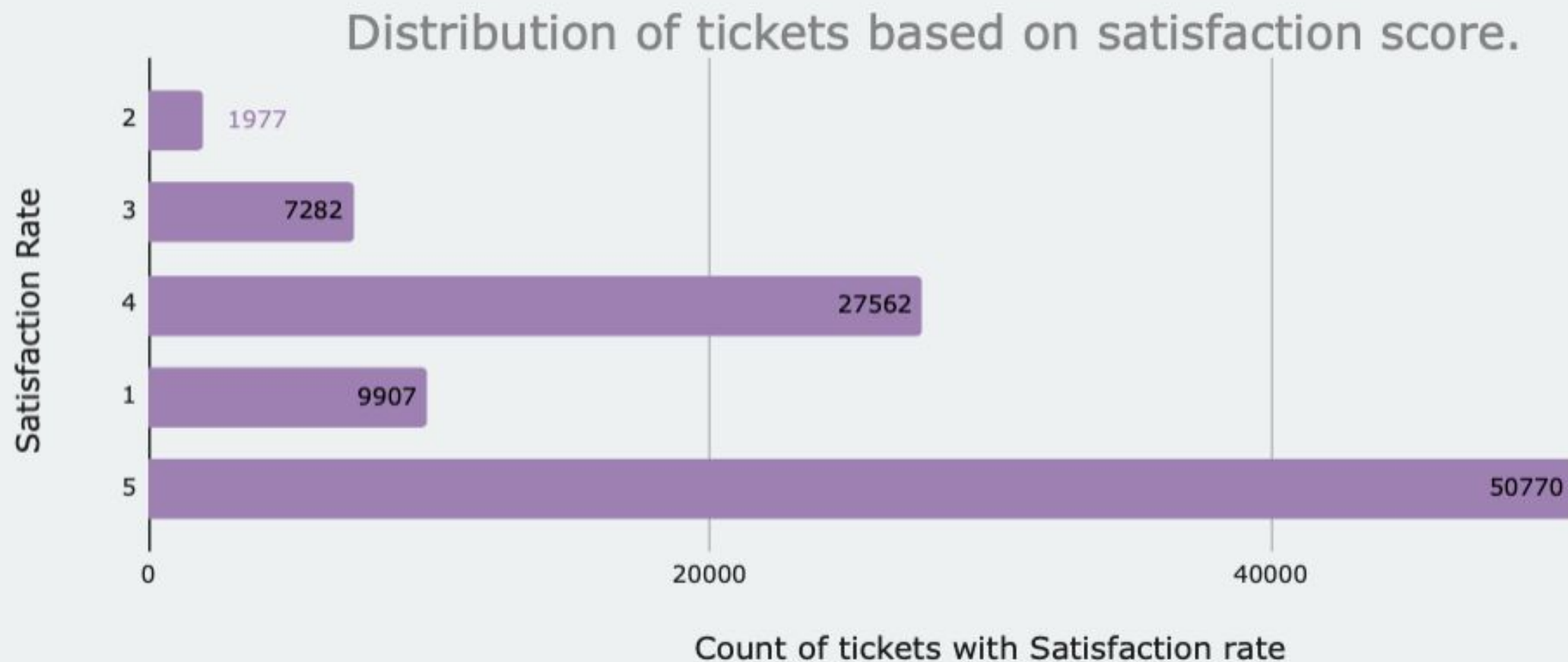
- Customer satisfaction is consistently high around **4.004to 4.22** for tickets handled by all IT agent age groups. This is a strong positive signal.
- The **29 - 33** age group of agents achieves the highest average satisfaction rate **4.22**.
- The **34 - 38** and **43 - 47** age groups of agents show slightly lower overall satisfaction scores compared to others.
- The **39 - 43** age group of agents, who have longer System and Software resolution times, also have a slightly lower satisfaction rate. This suggests that longer wait times, even if eventually resolved, can subtly impact

Tickets by Severity rate



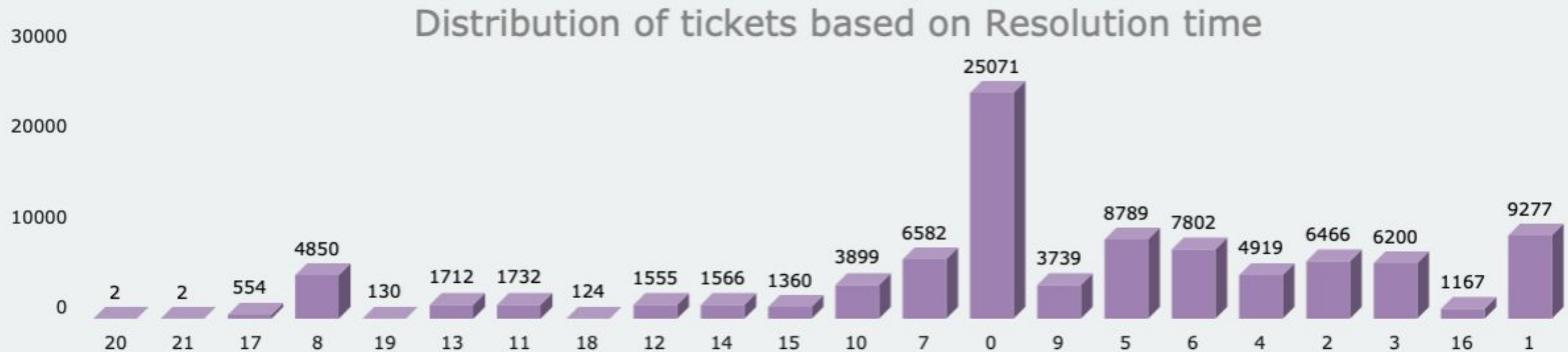
- Normal Severity dominates the ticket mix at 90.9%, so most of the workload comes from routine cases.
- Major, Minor, and Urgent tickets are much smaller shares at 5.5%, 2.3%, and 1.4%, meaning high-priority cases are rare.

Distribution of tickets based on satisfaction score



- Over 78% of tickets received ratings of 4 or 5, demonstrating high user confidence in IT support services and strong overall service quality.
- The relatively small number of low-rated tickets indicates that customer dissatisfaction is concentrated in a limited set of cases, making targeted process improvements more cost-effective than broad operational changes.
- Analyzing the causes of low-rated tickets can help eliminate recurring service issues and further increase customer retention and satisfaction.

Distribution of tickets over time



- The majority of tickets are resolved on the same day, demonstrating that operational processes are highly efficient for routine IT requests.
- The small proportion of long-resolution tickets likely represents complex Hardware and System issues, which consume disproportionate support resources and should be prioritized for process optimization.
- Reducing the number of high-resolution-time tickets will have the greatest impact on improving overall operational efficiency, reducing support costs, and increasing customer satisfaction.

Dashboard

IT TICKET ANALYSIS



Total Tickets

97,498



Avg Resolution Time

4.55



Avg Satisfaction Rate

4.1

Year All ▾

Priority All ▾

Severity All ▾

Issue Type All ▾

Request Categ... All ▾

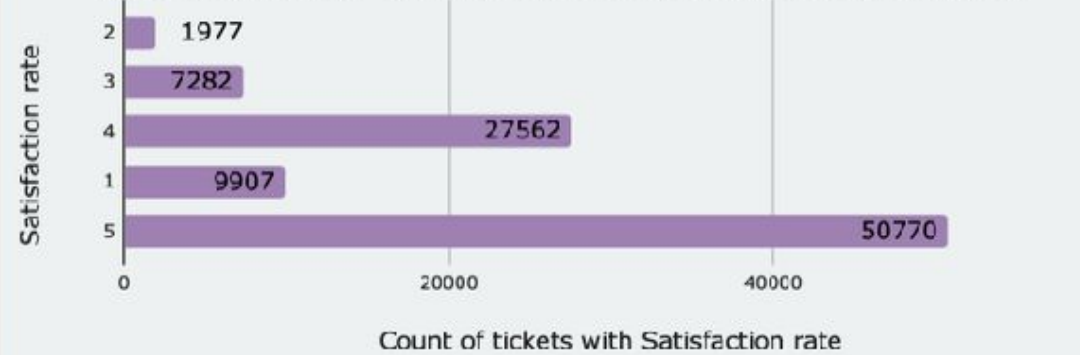
Count of ticket IDs by time.



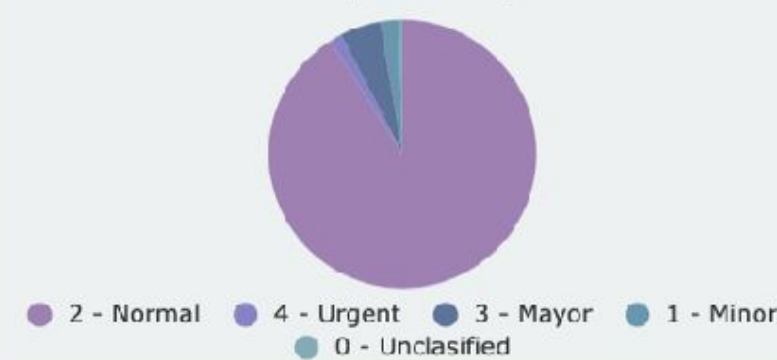
Distribution of tickets based on Resolution time



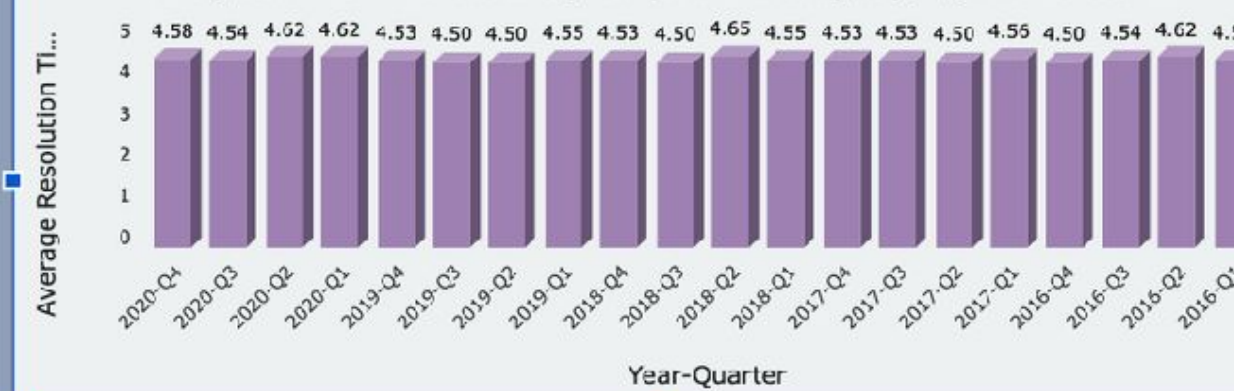
Distribution of tickets based on satisfaction score.



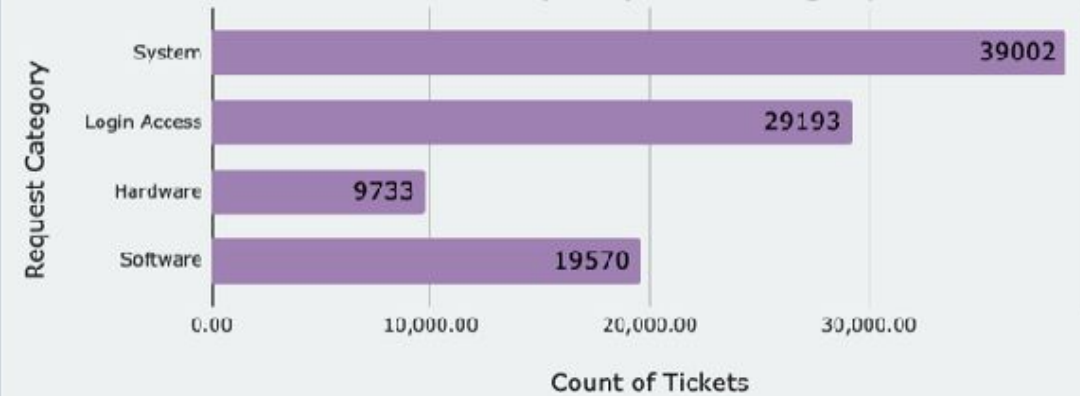
Ticket by severity rate



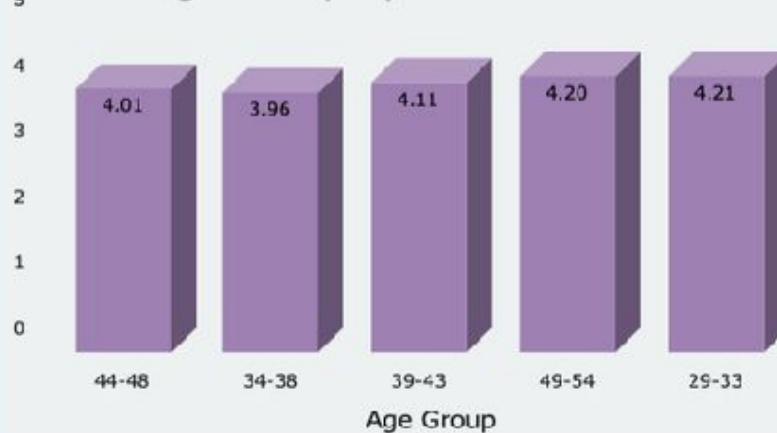
Avg Resolution time by request category, Quarter wise



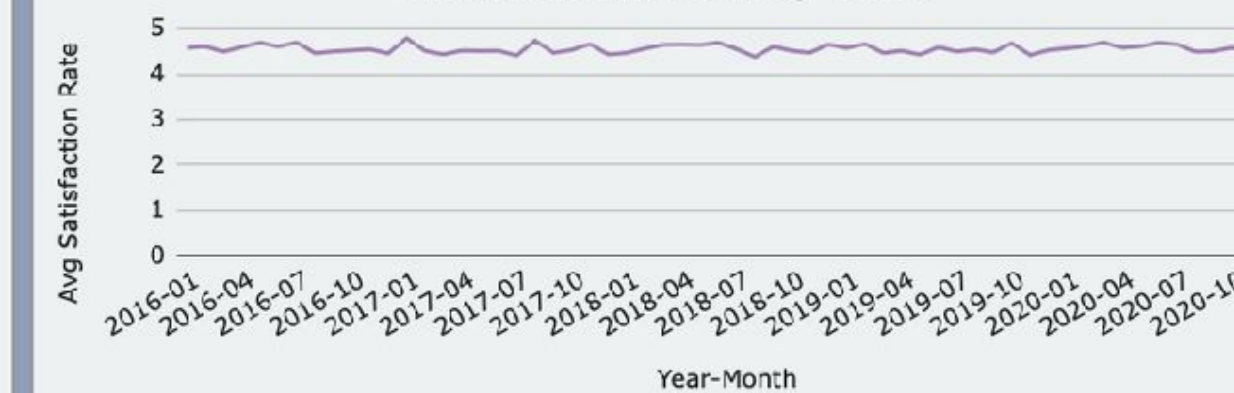
Count of tickets by request category.



Age Group by Resolution Time



Satisfaction Rate By Time



Strategic Reccomendations

1. **Upgrade the IT Ticket System (Top Priority):** Enable smart ticket routing, enforce priority tagging, and add self-service tools such as an AI chatbot and knowledge base to cut delays and boost employee experience.
2. **Optimize Workforce Performance:** Offer focused training and mentorship for low-performing agents to close gaps in resolution time and customer satisfaction across the team.
3. **Adopt a smart resource strategy:** invest in software first, expand the IT team later, and track KPIs regularly for continuous improvement.

Conclusion

Key Findings

- Hardware and System requests are the primary contributors to longer ticket resolution times.
- Automation significantly improves efficiency, as demonstrated by Login Access tickets.
- Differences in agent performance indicate opportunities for targeted training and process improvements.
- Technology investment offers the greatest potential for improving IT service performance.

Final Takeaway

- By combining technology upgrades, targeted workforce development, and data-driven decision-making, the organization can reduce resolution times, improve customer satisfaction, and build a more efficient and scalable IT support system.

Conclusion

- This analysis provides a clear roadmap to transform a growing but reactive IT support function into a proactive, efficient, and scalable operation. By strategically investing in our people and technology, we can sustain high customer satisfaction while successfully meeting the future demands of our organization.
- The IT support team is facing a challenge of ever-increasing ticket volume, but has successfully maintained high customer satisfaction. The current system is inefficient, with issues like unassigned priorities. The most effective short-term solution is to invest in training to empower agents and reduce resolution times. For long-term sustainability, a strategic upgrade of the ticket management software is essential to proactively reduce demand and ensure a scalable operation.

